Adaptive Multi-Frame Deferral for Promptable Video Segmentation

Anonymous CVPR submission

Abstract

Promptable video segmentation models such as SAM2 propagate an object mask through a video from a few expert prompts. The propagated mask can drift under occlusion and appearance change, requiring further expert intervention. Existing corrective frame-selection methods primarily determine where to intervene, either under fixed annotation budgets or through sequential corrections. However, they do not jointly determine how many frames to correct and which frames to select based on their combined video-wide segmentation benefit, while accounting for a uniform penalty per requested correction. We formulate corrective prompting as a variable-cardinality learning-to-defer problem, using a uniform cost per requested correction rather than measured annotation time or difficulty. Our framework determines whether to request assistance, how many corrections are justified, and which frames to correct. A cardinality model predicts the number of corrections, while a multi-frame selection model selects frames conditioned on those already selected. Experiments with SAM2.1 on four datasets spanning object transformation, camouflage, colonoscopy, and thyroid ultrasound demonstrate the best quality–cost trade-off among learned methods. Relative to initial propagation, our method improves mIoU by 5.6%–47.8%. *Compared with Top-*$k(x)$*, it reduces corrective prompts by* 3.1%–9.2% *and system cost by* 5.1%–10.7%, *while applying corrections in a single round. Code is released upon acceptance.* o

1. Introduction

Promptable video segmentation models such as SAM2 [14] propagate object masks through a video from a small number of mask, box, or point prompts. Yet propagation can deteriorate under appearance changes, occlusion, rapid motion, and background similarity [3], requiring corrective expert prompts [10]. Crucially, the amount of correction needed differs across videos: a reliable propagation may require none, whereas difficult sequences may benefit from several interventions. Since each requested corrective frame consumes expert effort, an effective system must decide *whether* to request corrections, *how many* to request, and *where* to place them. We study these decisions under a uniform per-correction cost, measuring intervention effort by the number of requested annotations rather than estimating frame-specific annotation time.

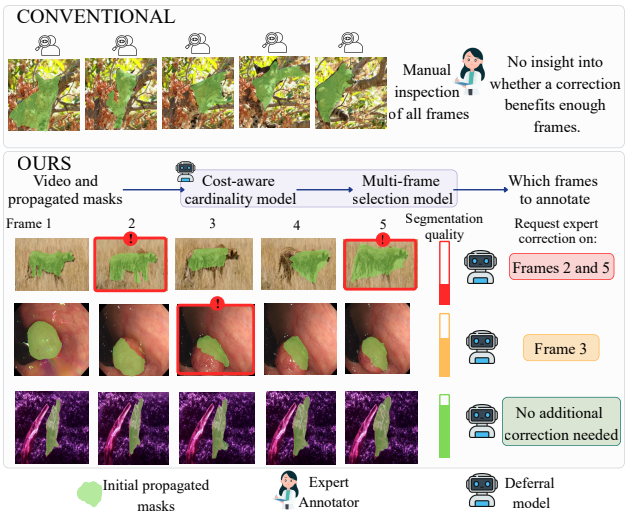


Figure 1. Corrective prompting requires deciding not only where to intervene, but also whether additional expert assistance is worthwhile and how many frames to correct. Under a uniform cost per corrective annotation, our framework selects a video-dependent correction cardinality and the corresponding frames. All selected corrections are requested together and incorporated through a single re-propagation, avoiding repeated correction rounds.

Existing corrective frame-selection methods primarily address *where* to annotate, using uncertainty, predicted mask quality, expected improvement, or diversity [2, 18, 22]. Some select frames under a prescribed annotation budget, while others decide on additional interventions sequentially, alternating frame selection, expert annotation, and mask propagation [18]. These strategies can identify informative frames, but they do not directly address the cost-sensitive choice of how many corrections to request *together* for a given video. A fixed number may spend unnecessary annotations on videos with reliable propagation or provide too little assistance for videos with multiple failures.

Learning to defer (L2D) offers a principled perspective on this decision by balancing predictive performance against the cost of consulting an expert [9, 16]. Beyond binary deferral, recent work considers how many experts to consult in classification [8, 23]. In promptable video segmentation, L2RP [13] learns whether to retain the initial propagation or request a correction, but is restricted to at most one corrective frame. Extending this decision to multiple frames introduces a distinct challenge: a correction affects masks throughout the video, so the value of requesting k annotations depends on the particular frames selected and their combined effect after propagation. Thus, deciding *how many* corrections are worthwhile is coupled with deciding *which* frames to correct.

We formulate this task as *cost-aware variable-cardinality deferral* for promptable video segmentation. Our framework first learns a context-conditioned multi-frame selector that constructs correction sets for different cardinalities while accounting for previously selected frames. A separate cardinality model is then trained for each specified unit correction cost, using the segmentation loss of the selected sets together with a penalty proportional to their cardinality. At inference, the corresponding trained model determines whether to retain the initial propagation or request k additional frames. The selected frames are annotated in one batch and used in a single corrected propagation, avoiding intermediate expert-interaction rounds.

Our main contributions are:

- We formulate corrective prompting as a cost-aware variable-cardinality deferral problem that determines whether, how many, and where corrective annotations should be requested under a uniform per-intervention cost.
- We develop a framework that learns cost-aware correction cardinality and constructs complementary correction sets through context-conditioned frame selection, enabling all corrections to be requested in a single interaction round.
- We evaluate the resulting segmentation-quality–cost trade-off on four diverse video datasets spanning object transformation, camouflage, colonoscopy, and thyroid ultrasound, against frame-selection and adapted learning-to-defer baselines.

2. Related work

2.1. Frame selection for corrective prompting

Frame selection for efficient human annotation has been studied in both active learning and interactive video segmentation. Active-learning approaches often prioritise frames using criteria such as uncertainty, representativeness, or diversity, with the acquired annotations subsequently used to improve the model [6, 12, 17]. For example, HVSA [11] combines uncertainty- and diversity-based selection with video-specific fine-tuning.

Interactive video segmentation considers a more directly related setting, in which user corrections are used to improve the propagated masks of the current video while the segmentation model remains fixed [20]. Within this setting, several methods have studied where and when the user should intervene. IVOS-W [22] recommends the next frame according to the expected video-level improvement after annotation, showing that the frame with the poorest current mask is not necessarily the most useful one to correct. EVA-VOS [2] selects the next frame using representations derived from the image and predicted mask, favouring frames that differ from those already annotated. Lazy-XMem [18] instead monitors uncertainty in the propagated masks and triggers a correction when the prediction becomes unreliable. Although these methods improve where the user intervenes, they select correction frames sequentially. When multiple corrections are needed for a video, this requires repeated cycles of propagation, expert review, and frame selection before all necessary corrections are identified [2]. Selecting a set of correction frames for each video can reduce this back-and-forth process while allowing the number of requested frames to adapt to the current propagation quality and the cost of expert assistance.

2.2. Learning to defer for promptable video segmentation

Learning to defer framework decides whether to retain a machine prediction or request expert assistance. Conventional formulations learn this decision by balancing the expected errors of the machine and expert against the cost of expert consultation [4, 7, 9, 16].

Related human–AI collaboration methods extend this idea to settings with multiple experts. LECODU [23] predicts how many experts should be consulted for each input and then randomly samples that number from the available expert pool before combining their predictions with the machine output. Top-k [8] learning to defer instead ranks experts according to their estimated suitability for the input and selects the highest-ranked k. Its adaptive Top-k(x) formulation additionally allows the consultation cardinality to vary across inputs.

Most L2D methods target classification. L2RP [13] is, to our knowledge, the first application of L2D to promptable video segmentation, but it defers at most once per video. Extending the setting to multiple corrective prompts is not straightforward, because each correction re-propagates through the video and changes how much the remaining candidates would add. This also prevents direct application of Top-$k(x)$ [8], which scores each expert independently and can therefore select several adjacent frames that address the same failure. Our framework instead selects the corrective frames jointly, conditioning each choice on the frames

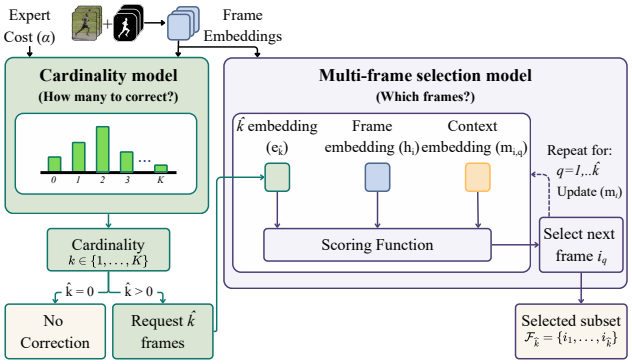


Figure 2. Overview of the proposed cost-aware multi-frame deferral framework. Given the video frame embeddings and expert cost α, the cardinality model predicts whether correction is required and, if so, the number of frames $\widehat{k}$ to prompt. Conditioned on the predicted cardinality, the multi-frame selection model constructs the correction set $\mathcal{F}_{\widehat{k}} = \{i_1, \ldots, i_{\widehat{k}}\}$ over $\widehat{k}$ selection steps, updating the context after each selected frame. The resulting frames are then requested for expert prompting in a single correction round.

already selected, and allows the number of corrections to vary with the video.

3. Method

Overview. A promptable video segmentation model requires an initial user prompt to identify the object of interest. Given this required prompt, the segmentation model propagates the corresponding object mask throughout the video to produce an initial segmentation. Our framework evaluates this propagation to determine whether additional corrective prompting is required, how many corrections should be requested, and where they should be placed. As shown in Figure 2, the framework consists of two main components: a cardinality model and a multi-frame selection model. The cardinality model selects $\widehat{k} \in \{0, \ldots, K\}$ by balancing the expected segmentation improvement from additional corrections against their expert prompting cost. If $\widehat{k} = 0$, the initial propagation is retained because the expected improvement does not justify the expert cost. Otherwise, the multi-frame selection model selects $\widehat{k}$ frames for expert annotation. The expert annotates all selected frames, and the segmentation model uses the resulting prompts together to update the segmentation throughout the video in a single propagation.

3.1. Preliminaries

3.1.1. Learning-to-Defer.

Conventional learning to defer considers a system composed of a machine predictor and a human expert. Given an input x, the machine produces a prediction $\widehat{y}^m$, while the expert can provide an alternative prediction $\widehat{y}^e$. A deferral policy $r(x) \in \{0, 1\}$ decides whether to retain the machine prediction or defer to the expert. Given the ground-truth target y, the resulting system loss is

$$\mathcal{L}_{\text{L2D}}(\mathsf{r}, x, y) = \mathbf{1}[\mathsf{r}(x) = 0]\, \ell_m(y, \widehat{y}^m) + \mathbf{1}[\mathsf{r}(x) = 1]\, [\ell_e(y, \widehat{y}^e) + \alpha]\,, \tag{1}$$

where $\ell_m(\cdot)$ is the loss incurred when retaining the machine prediction, $\ell_e(\cdot)$ is the loss associated with the expert prediction, and α denotes the cost of requesting expert input. The policy therefore learns to defer only when the expected reduction in prediction loss justifies the additional expert cost.

3.1.2. Promptable Video Segmentation Setting.

In our setting of promptable video segmentation, the machine prediction is the segmentation obtained by propagating an initial user prompt throughout a video. Expert assistance does not replace this prediction directly; instead, the expert provides corrective prompts on selected frames, which the segmentation model uses to update the video-wide segmentation.

Formally, let $v = (x_1, \ldots, x_N)$ denote a video containing N frames, with corresponding ground-truth segmentation masks $y = (y_1, \ldots, y_N)$. For each frame i, p_i denotes the expert-provided annotation that serves as the user prompt to the segmentation model. The set $\mathcal{F}_{\text{init}} \subseteq \{1, \ldots, N\}$ contains the indices of the initially prompted frames. These initial prompts identify the object of interest and are required to initialise the promptable segmentation model; they are therefore outside the corrective-prompting decision considered by our framework.

Given the initial prompts, a promptable video segmentation model s produces the initial propagated segmentation $\widehat{y} = s\,(v, \{p_i \mid i \in \mathcal{F}_{\text{init}}\})$. The propagation $\widehat{y}$ plays the role of the machine prediction in the learning-to-defer formulation. The set of unprompted frame indices eligible for corrective prompting is $\mathcal{E} = \{1, \ldots, N\} \setminus \mathcal{F}_{\text{init}}$.

3.2. Cost-aware multi-frame deferral

Given the video v and its initial propagation $\widehat{y}$, the deferral policy determines whether corrective prompts are required and, if so, how many should be requested. Let K denote the maximum number of additional frame prompts that may be requested for a video, and assume that $|\mathcal{E}| \geq K$. The policy selects $r(v, \widehat{y}) \in \{0, \ldots, K\}$, where $r(v, \widehat{y}) = 0$ retains the initial propagation and $r(v, \widehat{y}) = k > 0$ requests prompts on k additional frames.

For each $k > 0$, let $\mathcal{F}_k \subseteq \mathcal{E}$ denote the selected set of correction-frame indices, where $|\mathcal{F}_k| = k$. After obtaining the corresponding expert prompts, the resulting segmentation is $\widehat{y}_k = s\,(v, \{p_i \mid i \in \mathcal{F}_{\text{init}} \cup \mathcal{F}_k\})$.

Let $\ell(y, \widehat{y})$ denote the video-level segmentation loss. The

cost-aware multi-frame deferral objective is

$$\mathcal{L}_{\text{MF-L2D}} = \mathbf{1}[r(v,\widehat{y}) = 0]\,\ell(y,\widehat{y}) + \sum_{k=1}^{K} \mathbf{1}[r(v,\widehat{y}) = k]\,[\ell(y,\widehat{y}_k) + \alpha k]\,, \tag{2}$$

where α is the cost of one additional expert prompt. For $k > 0$, the reduction from $\ell(y,\widehat{y})$ to $\ell(y,\widehat{y}_k)$ represents the segmentation gain obtained from correction, while αk represents the corresponding expert cost.

We implement this decision using a cardinality model and a multi-frame selection model. The cardinality model determines whether the initial propagation should be retained or predicts the number of correction frames to request. Given a positive cardinality, the multi-frame selection model selects the corresponding frames from $\mathcal{E}$.

3.3. Cardinality model

The cardinality model implements the deferral policy r. Given the video and its initial propagation, it outputs $K+1$ scores and assigns a probability $\Pr_\phi(k \mid v,\widehat{y})$ to each action $k \in \{0,\dots,K\}$. The action $k = 0$ retains the initial propagation, whereas $k > 0$ specifies the number of frames to be selected by the multi-frame selection model.

At inference, the predicted cardinality is $\widehat{k} = r(v,\widehat{y}) = \arg\max_{k\in\{0,\dots,K\}} \Pr_\phi(k \mid v,\widehat{y})$. If $\widehat{k} = 0$, no further expert prompt is requested. Otherwise, the multi-frame selection model selects $\mathcal{F}_{\widehat{k}}$, and the expert annotations on these frames are applied together to update the segmentation throughout the video.

3.4. Multi-frame selection model

For a specified cardinality $k > 0$, the multi-frame selection model selects k frame indices from $\mathcal{E}$ over $q = 1,\dots,k$ selection steps. This sequential process allows each decision to account for previously selected frames, avoiding redundant choices without explicitly enumerating every possible subset. The sequentiality is over the selected-set size rather than video time, and all remaining eligible frames are considered at each step.

Throughout these selection steps, the segmentation remains $\widehat{y}$. Previously selected frames are added to the selection context, but they are not yet annotated, and the segmentation model is not re-propagated. Thus, all k frames are selected before their annotations are requested from the expert.

An encoder produces a representation $\mathbf{h}_i$ for every frame $i \in \{1,\dots,N\}$ using the video frame x_i and its initially propagated mask $\widehat{y}_i$. At selection step q, let $\mathcal{F}_{q-1} = \{i_1,\dots,i_{q-1}\}$ contain the frame indices selected at the preceding steps. The corresponding context is $\mathcal{C}_{q-1} = \mathcal{F}_{\text{init}} \cup \mathcal{F}_{q-1}$. At the first selection step ($q = 1$), $\mathcal{F}_0 = \varnothing$, and hence $\mathcal{C}_0 = \mathcal{F}_{\text{init}}$. This context allows each candidate to be evaluated by the information it adds to the frames already selected.

For each remaining candidate $i \in \mathcal{E} \setminus \mathcal{F}_{q-1}$, we apply soft cross-attention, using the candidate representation as the query and the context-frame representations as the keys and values. This produces a candidate-specific context representation $\mathbf{m}_{i,q} = \text{attn}\,(\mathbf{h}_i, \{\mathbf{h}_j \mid j \in \mathcal{C}_{q-1}\})$. Each supported cardinality k also has a learned embedding $\mathbf{e}_k$, allowing the scoring function to adapt its frame preferences to the requested set size. The candidate score is $z_{i,q}^{(k)} = \text{score}_\theta\,([\mathbf{h}_i; \mathbf{m}_{i,q}; \mathbf{e}_k])$. The attention and scoring functions are shared across all candidates, selection steps, and supported cardinalities.

The conditional probability of selecting candidate i at step q is $\Pr_\theta\,(i_q = i \mid v,\widehat{y},\mathcal{C}_{q-1},k) = \dfrac{\exp\left(z_{i,q}^{(k)}\right)}{\sum_{j\in\mathcal{E}\setminus\mathcal{F}_{q-1}} \exp\left(z_{j,q}^{(k)}\right)}$. During training, i_q is drawn by categorical sampling from this distribution. During inference, the candidate with the highest probability is selected. The partial set is then updated as $\mathcal{F}_q = \mathcal{F}_{q-1} \cup \{i_q\}$, and the context-dependent representations and candidate scores are recalculated for the next step. The original frame representations $\{\mathbf{h}_i\}_{i=1}^N$ remain fixed throughout this process.

After k steps, the sequence $\pi_k = (i_1,\dots,i_k)$ records the selection order, while the final output is the frame-index set $\mathcal{F}_k = \{i_1,\dots,i_k\} \subseteq \mathcal{E}$. The probability of the selection trajectory is

$$\Pr_\theta\,(\pi_k \mid v,\widehat{y},k) = \prod_{q=1}^{k} \Pr_\theta\,(i_q \mid v,\widehat{y},\mathcal{C}_{q-1},k)\,. \tag{3}$$

The sequence π_k is used to factorise the selection probability during training; the final decision is the unordered frame set $\mathcal{F}_k$.

3.5. Training objectives

Reward-Based Multi-Frame Selection. A video may contain multiple useful correction-frame sets, so the multi-frame selection model is not trained to reproduce a single predefined target set. Instead, it is trained using the segmentation loss obtained after prompting the selected frames.

For each cardinality k, we sample B alternative frame-index sets $\mathcal{R}_1,\dots,\mathcal{R}_B$ from $\mathcal{E}$, where $|\mathcal{R}_b| = k$ and $\mathcal{R}_b \neq \mathcal{F}_k$. Prompting the frames in $\mathcal{R}_b$ produces $\widehat{y}_{\mathcal{R}_b} = s\,(v,\{p_i \mid i \in \mathcal{F}_{\text{init}} \cup \mathcal{R}_b\})$. The lowest segmentation loss among these sampled alternatives is $b_k = \min_{1\le b\le B} \ell\,(y,\widehat{y}_{\mathcal{R}_b})$, and the advantage of the model-selected set is $a_k = b_k - \ell(y,\widehat{y}_k)$. The frame-selection loss for cardinality k is

$$\mathcal{L}_{\text{FS}}^{(k)} = -a_k \log \Pr_\theta\,(\pi_k \mid v,\widehat{y},k)\,. \tag{4}$$

Table 1. Comparison of variable-cardinality deferral across four datasets at $\alpha = 0.01$. We report segmentation quality (mIoU ↑), mean selected cardinality ($\widehat{k}$ ↓), and mean system cost ($c_{\widehat{k}}$ ↓), where $c_{\widehat{k}} = (1 - \text{mIoU}) + \alpha\widehat{k}$. Under $\mathcal{K}$, $\{0\}$, $\{0, 1\}$, and $\{0{:}4\}$ denote the allowed correction cardinalities. Rand., Single, Online, Indep., and Multi. denote random-subset selection, single-frame selection, online triggering, independent ranking, and multi-frame selection, respectively.

Method	k	Sel.	VOST			CAMotion			SUN-SEG			VTUS		
			mIoU↑	$\hat{k}$↓	c↓	mIoU↑	$\hat{k}$↓	c↓	mIoU↑	$\hat{k}$↓	c↓	mIoU↑	$\hat{k}$↓	c↓
No correction	0	–	.473	0.00	.527	.802	0.00	.198	.711	0.00	.289	.627	0.00	.373
Random	0:4	Rand.	$.587_{\pm.005}$	2.00	.433	$.819_{\pm.003}$	2.00	.201	$.762_{\pm.008}$	2.00	.258	$.694_{\pm.004}$	2.00	.326
L2RP [13]	0,1	Single	$.602_{\pm.002}$	0.76	.406	$.822_{\pm.002}$	0.47	.183	$.774_{\pm.006}$	0.52	.231	$.722_{\pm.001}$	0.88	.287
Lazy-XMem [18]	0:4	Online	.644	3.33	.390	.845	3.11	.186	.830	3.65	.207	.764	4.00	.276
LECODU [23]	0:4	Rand.	$.658_{\pm.001}$	3.21	.374	$.837_{\pm.002}$	1.28	.176	$.836_{\pm.004}$	2.01	.184	$.755_{\pm.002}$	3.26	.278
Top-$k(x)$ [8]	0:4	Indep.	$.662_{\pm.005}$	2.82	.366	$.838_{\pm.002}$	1.30	.175	$.842_{\pm.005}$	1.80	.176	$.795_{\pm.005}$	2.90	.234
Ours	**0:4**	**Multi.**	$\mathbf{.699}_{\pm.002}$	2.56	**.327**	$\mathbf{.847}_{\pm.001}$	1.19	**.165**	$\mathbf{.850}_{\pm.003}$	1.71	**.167**	$\mathbf{.819}_{\pm.002}$	2.81	**.209**
Oracle	*0:4*	*Best*	.714	2.86	.314	.851	1.10	.160	.874	1.84	.144	.831	2.87	.198

A positive advantage indicates that the selected set produces a lower segmentation loss than the sampled alternatives and therefore increases its selection probability. A negative advantage reduces its probability.

The multi-frame selection model is trained across all supported cardinalities $k \in \{1, \ldots, K\}$, with $\mathcal{L}_{\text{FS}} = \frac{1}{K}\sum_{k=1}^{K} \mathcal{L}_{\text{FS}}^{(k)}$. The frozen segmentation model is executed within the training loop to generate the post-correction masks and rewards. For each cardinality, it evaluates the selected set and B sampled alternatives, requiring $K(B + 1)$ online propagations per video update without enumerating all possible frame subsets.

Cost-Aware Cardinality Training. The multi-frame selection model is trained first and then frozen. During cardinality-model training, the frozen selection model produces one frame set $\mathcal{F}_k$ for each supported cardinality $k > 0$. For every cardinality $k \in \{0, \ldots, K\}$, we compute the corresponding system cost as

$$c_k = \begin{cases} \ell(y, \widehat{y}), & k = 0, \\ \ell(y, \widehat{y}_k) + \alpha k, & k > 0. \end{cases} \tag{5}$$

Thus, c_0 is the system cost of retaining the initial propagation, whereas c_k for $k > 0$ additionally includes the cost of requesting k expert prompts. The cardinality targets therefore reflect the trade-off between segmentation quality and expert effort.

To make the objective in Equation (2) trainable, we adapt a cost-weighted multiclass surrogate loss [1]. The action costs are normalised independently for each video to obtain $\widetilde{c}_k \in [0, 1]$, and the cardinality loss is

$$\mathcal{L}_{\text{card}} = -\sum_{k=0}^{K} (1 - \widetilde{c}_k) \log \Pr_{\phi}(k \mid v, \widehat{y}). \tag{6}$$

Actions with lower system cost receive larger weights, allowing the model to learn the relative quality–cost trade-off among all supported cardinalities.

During cardinality model training, the frozen segmentation model evaluates one selected set for each positive cardinality, requiring K online propagations per video update. Neither training stage enumerates the combinatorial set of possible frame subsets. At inference, the cardinality model predicts one $\widehat{k}$ and the selector chooses all $\widehat{k}$ frames without intermediate segmentation reruns. Once their expert prompts are available, SAM2 incorporates them through a single corrected propagation.

4. Experiments

We evaluate the proposed framework by posing three questions. **Q1**, Does the variable-cardinality deferral improve the trade-off between segmentation quality and expert cost? **Q2**, Does the proposed multi-frame selection model identify better correction frames than existing frame-selection methods under a fixed correction budget? **Q3**, Does the cardinality model allocate expert effort to videos that benefit most from additional corrections?

4.1. Experimental setup

Datasets and prompting protocol. We evaluate on VOST [15], CAMotion [21], SUN-SEG [5], and VTUS [19], using the official train/test splits. We sample ten uniformly spaced frames per video for prompting, with an initial ground-truth mask provided in the first frame to identify the object of interest. This sampling protocol enables tractable exhaustive oracle comparisons, while our framework operates without combinatorial search. We further evaluate larger candidate pools in 5.3. SAM2 propagates over the complete video, and training rewards and video-level mIoU are computed over all frames. Unless otherwise stated, corrective prompts are ground-truth masks, and

Table 2. Fixed-cardinality frame selection at $k = 3$. Rounds column shows 1 when all three frames are selected in a single round, 3 when one frame is selected per round.

Method	Rounds	VOST	CAMotion	SUN-SEG	VTUS
Random frames	1	0.613	0.831	0.787	0.727
Min-IoU frames	1	0.647	0.843	0.791	0.748
HVSA [11]	1	0.663	0.844	0.828	0.781
IVOS-W [22]	3	0.655	0.845	0.818	0.769
EVA-VOS [2]	3	0.667	0.848	0.825	0.786
Top-k [8]	1	0.670	0.847	0.829	0.772
Ours	**1**	**0.688**	**0.852**	**0.862**	**0.799**
Oracle	*1*	*0.712*	*0.859*	*0.875*	*0.828*

we allow up to $K = 4$ additional corrections, enabling the method to address multiple propagation failures while evaluating variable-cardinality decisions within a bounded correction budget. Additional details and results for other values of K are provided in the appendix.

Implementation details. We use a frozen SAM2.1 Hiera-Tiny model [14] as the promptable video segmentation model. For each sampled frame, we use embeddings produced by the SAM2 memory encoder, which jointly encode frame appearance and the corresponding predicted mask. These representations are processed by independently trained spatial-temporal encoders for the multi-frame selection and cardinality models, followed by MLP decision heads. The frame selector additionally uses candidate-to-context attention and learned cardinality embeddings.

The frame selector is trained for 400 epochs with $B = 4$ sampled alternatives, after which it is frozen and the cardinality model is trained for 200 epochs. Both models use AdamW (learning rate 10^{-5}, weight decay 10^{-4}, and batch size 16) and are trained on an NVIDIA GeForce RTX 5090 GPU. Unless otherwise stated, we use $\alpha = 0.01$ and train separate cardinality models for different expert-cost weights to evaluate the quality–cost trade-off. All learned methods are evaluated over three random seeds. Reported $\pm$ values denote the standard deviation across these runs. Additional implementation details are provided in the appendix.

Evaluation metrics. We report video-level mean intersection over union (mIoU), the mean number of requested corrections ($\bar{k}$), and the average system cost, defined as $1 - \text{mIoU} + \alpha\bar{k}$, where α is a fixed penalty assigned to each requested correction.

Evaluation protocols and comparison methods. Q1: How effectively does the proposed framework balance segmentation quality and expert correction cost? We evaluate variable-cardinality deferral under a maximum correction budget of $K = 4$, comparing against a no-correction reference, a random policy, and four baselines. Lazy-XMem [18] triggers corrections using per-frame uncertainty, while L2RP [13] supports only $k \in \{0, 1\}$. We additionally adapt two cost-aware learning-to-defer (L2D) approaches originally proposed for multiple-expert classification to our video frame-selection setting, treating candidate correction frames as expert actions. LECODU [23] predicts the number of corrections to request and selects the corresponding frames at random, while Adaptive Top-$k(x)$ [8] predicts the correction cardinality and selects frames according to independently estimated correction utilities. We also include an oracle upper bound that jointly selects the cardinality and correction frames to minimise system cost.

We further examine the quality–cost trade-off across different expert-cost weights α. To analyse deferral decisions, we evaluate mIoU across non-deferral coverage levels by varying the deferral threshold. Finally, we analyse the adaptive allocation of corrections by grouping videos according to their initial segmentation mIoU and examining the mean selected cardinality and resulting mIoU improvement across different quality levels.

Q2: How effective is the proposed multi-frame selection model under a fixed correction budget? To evaluate frame selection independently of cardinality decisions, we fix the number of corrections per video and compare against random selection, a minimum-IoU heuristic that selects frames with the lowest IoU between the initial propagation and ground truth, a fixed-cardinality variant of Top-$k(x)$ [8] that selects the k frames with the highest independently estimated correction utilities, and adaptations of HVSA [11], EVA-VOS [2], and IVOS-W [22]. HVSA and Top-$k(x)$ select correction frames in a batch, whereas EVA-VOS and IVOS-W select frames sequentially, re-propagating after each correction. Our method selects all correction frames in a single batch without intermediate re-propagation. We also report an oracle upper bound that selects the frame subset yielding the highest mIoU for each fixed k.

We additionally present ablation studies, robustness evaluations, and qualitative results. Further details of the baseline adaptations, evaluation protocols, and additional analyses are provided in the appendix.

5. Results and Analysis

5.1. Cost-Aware Variable-Cardinality Deferral

Table 1 shows that, without correction, the initial propagation yields the lowest mIoU, while random correction improves it. L2RP requests the fewest corrections but is limited to a single frame, whereas Lazy-XMem requests 3–4 corrections. LECODU and Top-$k(x)$ incur similar system costs on CAMotion, while Top-$k(x)$ achieves lower costs

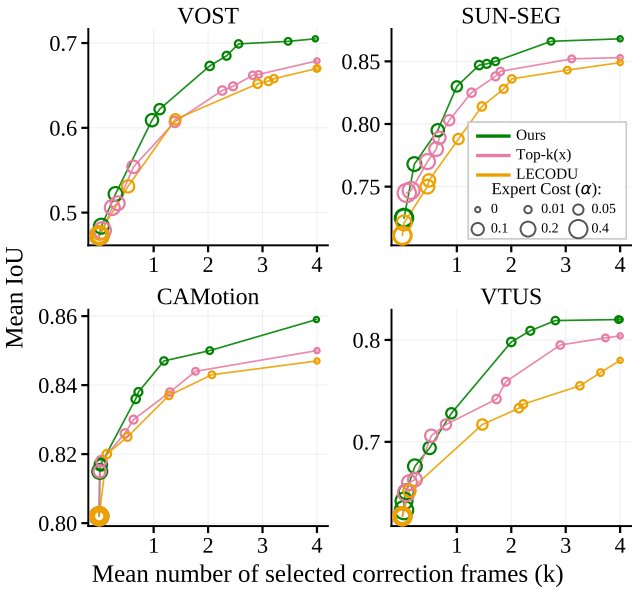


Figure 3. Quality–cost trade-off under varying expert-cost weight α. Each point reports the video-level mIoU and mean selected cardinality for one value of α, with marker size increasing with α. Our method is generally above the baselines at comparable correction budgets.

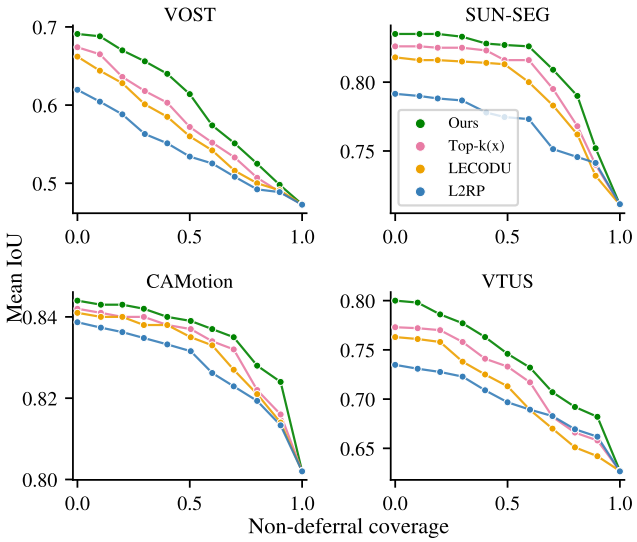


Figure 4. Mean IoU at different non-deferral coverage levels. Coverage is the fraction of videos that retain the initial segmentation without corrective prompting: 0 means that all videos are corrected, whereas 1 means that none are corrected. Higher curves indicate better segmentation quality at the same coverage.

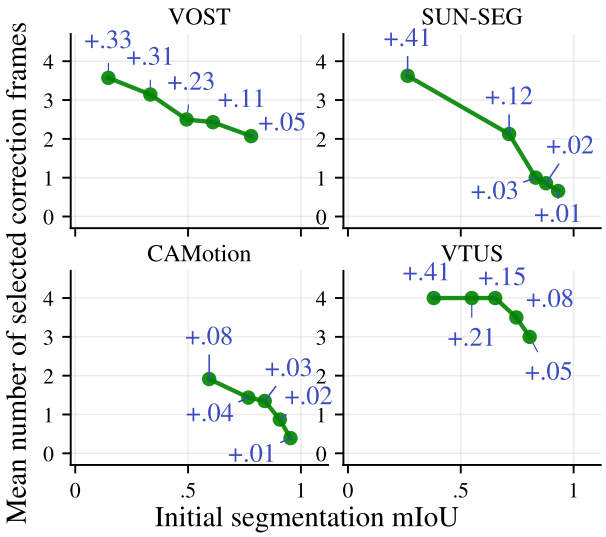


Figure 5. Mean number of selected correction frames versus initial segmentation quality. Videos are grouped into five initial-mIoU bins, and the labels show the mean mIoU gain after correction. Lower-quality segmentations receive more corrections and larger gains, indicating adaptive allocation of expert effort.

on VOST, SUN-SEG and VTUS. Our method achieves the highest mIoU with the lowest system cost among non-oracle methods. Compared with Top-$k(x)$, ours reduces the mean cardinality by 3.1–9.2% and the system cost by 5.4–11.4%. Although the oracle defines the upper bound, ours consistently remains the closest, reducing 28.1–75.0% of the system-cost gap between Top-$k(x)$ and the oracle.

Figure 3 shows that increasing α generally reduces both the mean selected cardinality and mIoU. Our method generally achieves higher mIoU than the baselines at comparable correction budgets, demonstrating a consistent quality–cost trade-off across different expert-cost settings.

Figure 4 further shows that our method achieves the highest mIoU across non-deferral coverage levels, indicating that its performance gains are consistent across a wide range of operating points rather than tied to a particular decision threshold.

Figure 5 shows that the policy allocates larger correction budgets to videos with lower initial mIoU and fewer corrections to those with higher initial mIoU. The mIoU gains are positive across all bins and are generally larger for lower-quality initial propagations. These results indicate that the policy adaptively allocates corrections according to the initial segmentation quality and the observed benefit of additional interventions.

5.2. Frame Selection at Fixed Cardinality

Table 2 compares frame-selection methods under a fixed correction budget of $k = 3$. IVOS-W and EVA-VOS perform sequential frame selection, re-propagating after each correction, whereas batch methods select all correc-

Table 3. Ablation of cardinality decisions on VTUS. We compare random and learned cardinality selection using the same multi-frame selector. Results are reported in video-level mIoU.

Cardinality selection	VTUS mIoU
Random	0.746
Learned (Ours)	**0.819**

tion frames in a single round from the initial segmentation. Despite not benefiting from intermediate propagation updates, our method achieves the highest non-oracle mIoU across all four datasets, improving over EVA-VOS by 0.5–4.5 percentage points and batch Top-k by 0.6–4.0 percentage points. These results demonstrate the effectiveness of our multi-frame selection model under a fixed correction budget. Additional results for different values of k are provided in the appendix.

5.3. Ablation Studies and Further Analysis

Cardinality decisions. Table 3 evaluates the contribution of learned cardinality decisions using the same multi-frame selector. On VTUS, learned cardinality achieves an mIoU of 0.819, compared with 0.746 for random cardinality selection, demonstrating the benefit of learning how many corrections to request for each video.

Cardinality and frame-selection components. Table 4 evaluates the contributions of learned cardinality decisions and selection-conditioned frame selection across all four datasets. At fixed $k = 4$, selection-conditioned ranking consistently outperforms static-context ranking, demonstrating the benefit of accounting for previously selected correction frames. The same improvement is observed under learned cardinality. Furthermore, learned cardinality reduces the number of requested corrections and system cost compared with fixed cardinality. Combining learned cardinality with selection-conditioned frame selection achieves the lowest system cost across all four datasets, demonstrating the benefits of both components.

Number of sampled frame subsets. Table 5 examines the effect of the number of sampled alternative frame sets B used during training. Increasing B generally improves segmentation performance. From $B = 1$ to $B = 4$, mIoU increases by 0.023 on VOST, 0.017 on CAMotion, and 0.019 on VTUS. SUN-SEG achieves its highest mIoU at $B = 2$, with comparable performance at $B = 4$. We therefore use $B = 4$ in our experiments.

Robustness to imperfect corrective prompts.. Table 6 evaluates the robustness of our framework to different prompting configurations on VTUS. We consider simulated point prompts with 1, 3, and 5 clicks per prompted frame, alongside ground-truth mask prompts. The same prompting configuration is used for initialization and subsequent corrections. Our method consistently outperforms Top-$k(x)$ across all prompting configurations, demonstrating its robustness to imperfect corrective prompts.

Candidate sampling density. Table 7 evaluates the effect of candidate sampling density on VTUS. We vary the number of uniformly sampled frames from 10 to 30 while keeping the maximum correction cardinality fixed at $K = 4$. Increasing the sampling density improves our video-level mIoU from 0.819 to 0.834, while reducing the mean selected cardinality from 2.81 to 2.76 and system cost from 0.209 to 0.194. Our method consistently achieves higher mIoU and lower system cost than Top-$k(x)$ across all sampling densities, demonstrating its effectiveness with larger candidate pools and finer temporal sampling.

Generalization across VOS models. Table 8 evaluates the generalization of our framework across three video object segmentation models: SAM2.1, Cutie, and SAMURAI on VTUS. Our method consistently achieves higher mIoU and lower system cost than Top-$k(x)$ across all three models, demonstrating that its effectiveness is not limited to the SAM2.1 segmentation model used in the main experiments.

Multiple correction rounds. Although our framework requests corrections in a single batch, the same policy can be reapplied after each re-propagation. Figure 6 evaluates this setting on SUN-SEG for maximum cardinalities $K \in \{1, 2, 3, 4\}$. Larger correction batches achieve a given mIoU in fewer interaction rounds than sequential methods, reducing the number of times expert intervention is requested. These results demonstrate that our framework can support iterative correction while retaining the benefits of batch frame selection.

5.4. Qualitative Results

Figure 11 presents qualitative comparisons on SUN-SEG for videos with different levels of initial segmentation quality. In the low-initial-IoU example, our method selects three corrective frames and achieves higher final IoU with lower system cost than Top-$k(x)$, which requests four corrections. In the high-initial-IoU example, our method correctly retains the initial propagation without additional corrections, whereas Top-$k(x)$ requests an unnecessary correction that reduces segmentation quality. These examples illustrate the ability of our framework to adapt correction decisions to the needs of individual videos.

6. Conclusion

We formulated corrective prompting for promptable video segmentation as a variable-cardinality learning-to-defer problem that accounts for how the value of each correction depends on the other selected frames. Our framework couples a cardinality model that determines whether and how many corrections are warranted under a given expert cost with a multi-frame selection model that conditions each

Table 4. Component ablation across four datasets at $\alpha = 0.01$. We compare fixed ($k = 4$) and learned cardinality with static-context and selection-conditioned frame selection. Static-context ranking scores candidate frames without conditioning on previously selected frames, whereas selection-conditioned ranking updates candidate scores after each selection. Both variants use matched architectures and training objectives.

		VOST			CAMotion			SUN-SEG			VTUS		
Cardinality	**Frame selection**	mIoU↑	$\hat{k}$	$c\downarrow$	mIoU↑	$\hat{k}$	$c\downarrow$	mIoU↑	$\hat{k}$	$c\downarrow$	mIoU↑	$\hat{k}$	$c\downarrow$
Fixed $k = 4$	Static-context	0.690	4.00	0.350	0.845	4.00	0.195	0.846	4.00	0.194	0.805	4.00	0.235
Fixed $k = 4$	Selection-conditioned	0.710	4.00	0.330	0.854	4.00	0.186	0.862	4.00	0.178	0.820	4.00	0.220
Learned k	Static-context	0.675	2.72	0.352	0.836	1.28	0.177	0.834	1.85	0.185	0.799	2.90	0.230
Learned k	**Selection-conditioned**	**0.699**	**2.56**	**0.327**	**0.847**	**1.19**	**0.165**	**0.850**	**1.71**	**0.167**	**0.819**	**2.81**	**0.209**

Table 5. Ablation on the number of sampled frame subsets B used to construct the reward baseline during training. Values are video-level mIoU.

Dataset	**B = 1**	**B = 2**	**B = 4**
VOST	0.676	0.681	**0.699**
CAMotion	0.830	0.832	**0.847**
SUN-SEG	0.836	**0.851**	0.850
VTUS	0.800	0.802	**0.819**

Table 6. Robustness to simulated point prompts on VTUS. We evaluate point-based prompting with 1, 3, and 5 clicks per prompted frame, compared with ground-truth mask prompts. The same prompting configuration is used for initialization and subsequent corrections. Results are reported in video-level mIoU.

Prompt type	**No correction**	**Top-$k(x)$**	**Ours**
1 click	0.495	0.634	**0.662**
3 clicks	0.552	0.715	**0.740**
5 clicks	0.588	0.755	**0.775**
Mask	0.627	0.795	**0.819**

Table 7. Effect of candidate sampling density on VTUS. We vary the number of uniformly sampled frames while keeping the maximum correction cardinality fixed at $K = 4$. We report video-level mIoU, mean selected cardinality, and system cost.

N	**Method**	**mIoU ↑**	**Mean k**	**Cost ↓**
10	Top-$k(x)$	0.795	2.90	0.234
10	**Ours**	**0.819**	**2.81**	**0.209**
20	Top-$k(x)$	0.802	2.87	0.227
20	**Ours**	**0.828**	**2.79**	**0.200**
30	Top-$k(x)$	0.806	2.85	0.223
30	**Ours**	**0.834**	**2.76**	**0.194**

choice on the frames already selected and requests the resulting corrections in a single round. Experiments with a SAM2 model across four visual and clinical datasets show that our method achieves the highest mIoU and lowest system cost among non-oracle methods while requesting fewer corrections than the strongest baseline. These findings show that coupling cost-aware cardinality decisions with multi-frame selection improves the quality–cost trade-off while consolidating corrective prompting into a single round.

Table 8. Generalization across video object segmentation models on VTUS. We evaluate the proposed framework with SAM2.1, Cutie, and SAMURAI, reporting video-level mIoU, mean selected cardinality, and system cost.

Method	**mIoU ↑**	**Mean k**	**Cost ↓**
SAM2.1 Hiera-Tiny			
No correction	0.627	0.00	0.373
Top-$k(x)$	0.795	2.90	0.234
Ours	**0.819**	**2.81**	**0.209**
Cutie			
No correction	0.647	0.00	0.353
Top-$k(x)$	0.780	2.88	0.249
Ours	**0.802**	**2.76**	**0.226**
SAMURAI			
No correction	0.665	0.00	0.335
Top-$k(x)$	0.793	2.70	0.234
Ours	**0.812**	**2.59**	**0.214**

LLM Usage Statement

OpenAI Codex was used to assist with migrating the existing manuscript and supplementary material from the AAAI template to the CVPR template, adapting LaTeX formatting, and checking compilation and layout.

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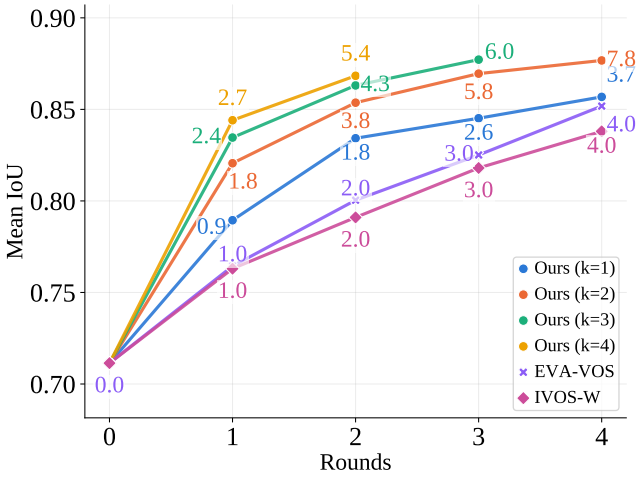


Figure 6. Mean IoU across correction rounds on SUN-SEG at $\alpha = 0.001$. Our framework selects correction frames in batches with maximum cardinality $K \in \{1, 2, 3, 4\}$, whereas EVA-VOS and IVOS-W select one frame per round. After each round, the video is re-propagated. Point labels indicate the cumulative number of corrections.

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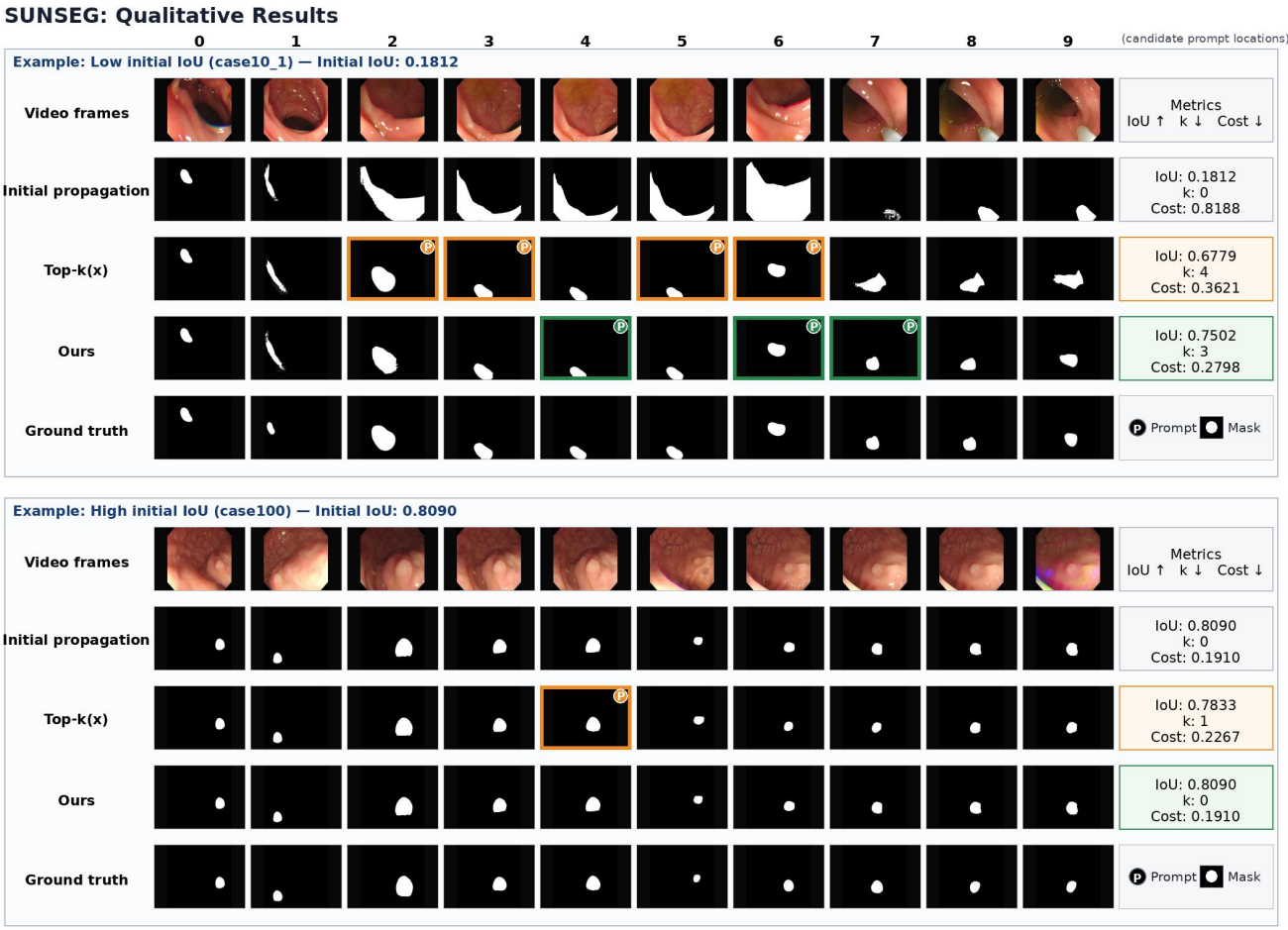


Figure 7. Qualitative comparison on SUN-SEG for videos with low and high initial segmentation quality. Our method selects fewer corrective frames while achieving higher final IoU and lower system cost than Top-$k(x)$. In the high-initial-IoU example, our method avoids an unnecessary correction. Coloured borders and prompt markers indicate the selected correction frames.

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Adaptive Multi-Frame Deferral for Promptable Video Segmentation

Supplementary Material

A. Datasets

We evaluate the proposed method on four datasets covering substantially different video-segmentation settings: object transformation in VOST, camouflaged objects in CAMotion, colonoscopy in SUN-SEG, and thyroid ultrasound in VTUS. This diversity allows us to assess frame selection under different appearance changes, object visibility, imaging modalities, and mask-propagation failures.

We preserve the original training and validation splits of each dataset. To increase the number of training samples, long training videos are divided into shorter overlapping clips, while VOST videos are used without division. Validation videos are kept unchanged. The dataset sizes and clip-generation settings are reported in Table 9.

Table 9. Dataset splits and temporal partitioning used to construct the training clips. Window and stride are measured in frames. Dashes indicate that the original videos are used without partitioning.

Dataset	Original		Partitioning		Final	
	Train	Val.	Window	Stride	Train	Val.
VOST	572	71	–	–	572	71
SUN-SEG	112	37	100	10	917	37
VTUS	70	30	50	5	686	30
CAMOtion	359	115	100	10	528	115

For each training clip and validation video, we define ten uniformly spaced temporal locations. The first receives the initial ground-truth mask prompt, and the remaining nine are correction candidates. These locations only determine where prompts can be placed; SAM2 propagates over the entire sequence, and mIoU is computed over all frames in that sequence.

B. Additional implementation details

B.1. Promptable segmentation model configurations

We use a frozen SAM2.1 Hiera-Tiny model with an internal image size of 512×512 to reduce propagation time without a noticeable loss in segmentation accuracy.

We use ground-truth masks as the initial and corrective prompts. A ground-truth mask represents an ideal expert correction after the system defers a frame for annotation. This removes variation caused by prompt quality and allows the experiments to isolate the decisions of whether to request correction, how many frames to correct, and which frames to select.

B.2. Architectural details

Feature representation and encoder. For each sampled frame location, we use the mask-memory feature map produced by the frozen SAM2.1 memory encoder. This representation jointly encodes the frame appearance and its current predicted mask, and is used by SAM2 to condition subsequent mask propagation.

Each mask-memory feature map is adaptively average-pooled to a 4×4 grid, yielding 16 spatial tokens with 64 channels. The tokens are projected to 128 dimensions, combined with learned spatial positional embeddings, and processed by one four-head spatial Transformer layer. Spatial mean pooling then produces one representation per frame. After adding learned temporal positional embeddings, two four-head temporal Transformer layers exchange information across the sampled video sequence, producing a contextualized 128-dimensional representation $\mathbf{h}_i$ for each location. The frame selector and cardinality model use independently trained encoders with this architecture and do not share parameters.

Multi-frame selection head. The selector uses single-head soft cross-attention, with each candidate frame as the query and the prompted and previously selected frames as keys and values. The resulting 128-dimensional context vector is concatenated with the 128-dimensional candidate representation and a learned 128-dimensional embedding of the requested cardinality. The resulting 384-dimensional vector is processed by layer normalization and a two-layer MLP with dimensions $384 \to 128 \to 1$, using GELU activation and dropout 0.1. Previously prompted and selected frames are masked. The next frame is sampled categorically during training and selected greedily during inference.

Cardinality head. The cardinality model mean-pools its encoded frame representations to obtain a 128-dimensional video-level vector. This vector is processed by layer normalization and a two-layer MLP with dimensions $128 \to 128 \to (K+1)$, using GELU activation and dropout 0.1. The outputs represent non-deferral and cardinalities $1, \ldots, K$. The frame selector is frozen during cardinality-model training.

The complete deferral decision, including cardinality prediction and frame selection, takes approximately 30 ms per video, excluding SAM2 propagation.

Table 10. Variable-cardinality frame selection under maximum correction budgets $K_{\max} \in \{2, 3, 4, 5, 6\}$ at $\alpha = 0.01$. For each method and dataset, we report video-level segmentation quality (mIoU ↑) and the mean number of selected correction frames ($\bar{k}$ ↓). Boldface denotes our method.

		VOST		CAMOtion		SUN-SEG		VTUS	
K	**Method**	mIoU	$\bar{k}$	mIoU	$\bar{k}$	mIoU	$\bar{k}$	mIoU	$\bar{k}$
2	LECODU	0.622	2.00	0.828	1.01	0.802	2.00	0.739	2.00
2	Top-k (x)	0.638	1.57	0.831	0.74	0.813	1.32	0.756	1.80
2	**Ours**	**0.671**	**2.00**	**0.838**	**0.93**	**0.830**	**1.29**	**0.777**	**1.97**
3	LECODU	0.642	3.00	0.830	0.95	0.814	2.12	0.758	3.00
3	Top-k (x)	0.668	2.33	0.832	0.89	0.827	1.59	0.774	2.37
3	**Ours**	**0.686**	**2.44**	**0.841**	**1.01**	**0.841**	**1.60**	**0.797**	**2.73**
4	LECODU	0.658	3.21	0.837	1.28	0.836	2.01	0.755	3.26
4	Top-k (x)	0.662	2.82	0.838	1.30	0.842	1.80	0.795	2.90
4	**Ours**	**0.699**	**2.56**	**0.847**	**1.19**	**0.850**	**1.71**	**0.819**	**2.81**
5	LECODU	0.670	4.63	0.841	1.36	0.847	2.52	0.803	3.86
5	Top-k (x)	0.671	3.50	0.840	0.94	0.845	2.05	0.804	3.00
5	**Ours**	**0.706**	**3.73**	**0.849**	**1.13**	**0.866**	**2.24**	**0.837**	**4.20**
6	LECODU	0.684	4.96	0.849	1.22	0.855	2.26	0.819	4.42
6	Top-k (x)	0.680	4.09	0.842	1.33	0.852	2.24	0.811	4.47
6	**Ours**	**0.715**	**3.67**	**0.855**	**1.45**	**0.877**	**2.32**	**0.847**	**4.07**

C. Additional baseline details

We adapt baselines from learning to defer and video frame selection. All baselines use the same frozen SAM2.1 Hiera-Tiny model, sampled prompt locations, and ground-truth mask prompts. The first location provides the initial prompt, while the remaining locations are eligible for additional expert prompts. The methods differ in how they determine the prompt cardinality k and select the corresponding frames.

C.1. Reference policies.

The non-deferral policy requests no additional prompt ($k = 0$) and retains the initial SAM2 propagation. For a given $k > 0$, the minimum-IoU policy selects the k eligible frames with the lowest frame-level IoU. This approximates an expert visually inspecting the propagation and prompting the frames that appear most erroneous.

For fixed-cardinality evaluation, k is specified in advance. The random policy uniformly samples a k-frame subset without replacement, while the oracle selects the subset of size k that produces the highest video-level mIoU. For variable-cardinality evaluation, the random policy first samples k uniformly from $\{0, \ldots, K\}$ and then samples k eligible frames. The oracle jointly selects the cardinality and frame subset with the lowest cost $1 - \mathrm{mIoU} + \alpha k$. In both settings, the oracle is computed by evaluating all eligible combinations only to compare our method with the best achievable performance; this exhaustive search is not required during training or inference.

C.2. Learning-to-defer adaptations.

All learning-to-defer baselines use the same inputs and cardinality architecture as our method. Only their output spaces and training losses are changed to follow the corresponding original formulations.

LECODU and Adaptive Top-$k(x)$ were originally formulated for classification with multiple users or experts. We treat each eligible frame location as a deferral action, where selecting it requests an expert mask prompt. In our LECODU adaptation, the predicted number of users determines k, after which k eligible frames are sampled uniformly. Adaptive Top-$k(x)$ predicts k, estimates the utility of prompting each frame independently, and selects the k highest-ranked frames.

L2RP was originally designed for promptable video segmentation. We adapt it to our inputs and cardinality architecture while retaining its original loss and output space. It therefore decides between retaining the current propagation and requesting one expert prompt at a selected frame, supporting only $k \in \{0, 1\}$.

C.3. Video frame-selection adaptations.

Lazy-XMem, HVSA, EVA-VOS, and IVOS-W use different criteria for determining when or where annotations should be requested. We retain their triggering or frame-selection procedures while using the same frozen SAM2.1 model and eligible frame locations. Lazy-XMem uses prediction uncertainty to trigger a prompt; HVSA selects frames using uncertainty and diversity; EVA-VOS uses its

Table 11. Fixed-cardinality frame selection for $k \in \{2, 3, 4\}$. Batch methods select all k frames before a single propagation, whereas sequential methods select and propagate one correction at a time. We report video-level mIoU; the oracle gives the best frame subset for each k.

		VOST			CAMotion			SUN-SEG			VTUS		
Method	**Selection**	k=2	k=3	k=4	k=2	k=3	k=4	k=2	k=3	k=4	k=2	k=3	k=4
Random frames	Batch	0.584	0.613	0.632	0.825	0.831	0.845	0.766	0.787	0.802	0.693	0.727	0.760
Worst-IoU frames	Batch	0.591	0.647	0.663	0.836	0.843	0.842	0.787	0.791	0.818	0.721	0.748	0.772
HSVA (2023)	Batch	0.635	0.663	0.679	0.837	0.844	0.845	0.792	0.828	0.830	0.770	0.781	0.794
IVOS-W (2021)	Sequential	0.606	0.655	0.675	0.838	0.845	0.848	0.791	0.818	0.838	0.720	0.769	0.796
EVA-VOS (2024)	Sequential	0.629	0.667	0.681	0.846	0.848	0.854	0.800	0.825	0.852	0.763	0.786	0.810
Top-k (2025)	Batch	0.643	0.670	0.683	0.844	0.847	0.850	0.811	0.829	0.845	0.750	0.772	0.804
Ours	**Batch**	**0.671**	**0.688**	**0.701**	**0.849**	**0.852**	**0.859**	**0.844**	**0.862**	**0.868**	**0.787**	**0.799**	**0.820**
Oracle	*Batch*	*0.695*	*0.712*	*0.721*	*0.854*	*0.859*	*0.862*	*0.860*	*0.875*	*0.881*	*0.813*	*0.828*	*0.835*

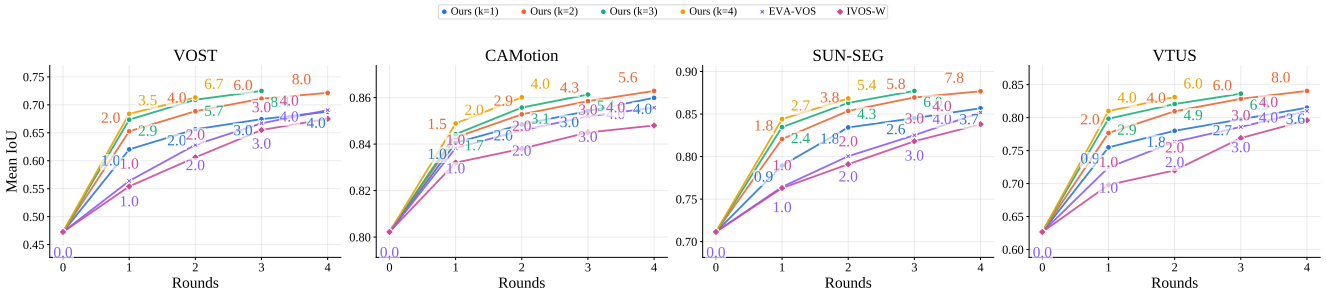


Figure 8. Mean IoU over corrective-prompting rounds on all four datasets at $\alpha = 0.001$. Our models with maximum cardinality $K \in \{1, 2, 3, 4\}$ select multiple correction frames as a batch in each round, whereas EVA-VOS and IVOS-W select one frame per round. After each round, the selected frames are prompted and the full video is re-propagated. Point labels indicate the mean cumulative number of selected correction frames.

frame-selection component; and IVOS-W uses its learned frame-recommendation policy.

The original HVSA framework performs video-specific test-time fine-tuning after acquiring annotations. Our adaptation uses only its frame-selection procedure and does not retrain the segmentation model. Instead, selected frames receive ground-truth mask prompts and the frozen SAM2.1 model propagates the updated segmentation. Further, sequential methods re-propagate after each prompt before selecting the next frame, whereas batch methods select all k frames from the current propagation and apply their prompts together.

D. Additional results

D.1. Variable cardinality for different maximum budgets

The main experiments use $K = 4$ additional corrections. Table 10 extends this analysis to $K_{\max} \in \{2, 3, 4, 5, 6\}$ at $\alpha = 0.01$. Our method achieves the highest mIoU across all datasets and correction limits.

D.2. Fixed-cardinality frame selection for different budgets

Table 11 shows that our method achieves the highest non-oracle mIoU across all datasets and $k \in \{2, 3, 4\}$. The gains are particularly clear on VOST, SUN-SEG, and VTUS, while competing methods perform more similarly on CAMotion. This indicates that our selector identifies more effective sets of correction frames under the same fixed expert budget.

D.3. Iterative application across datasets

Figure 8 extends the SUN-SEG analysis in the main paper to all four datasets. We use $\alpha = 0.001$ so that the policy continues to request useful corrections over multiple rounds, rather than quickly selecting $k = 0$ as segmentation quality improves. This setting also provides a closer comparison with EVA-VOS and IVOS-W, which focus on frame selection without accounting for expert cost. The results show that batch selection can obtain gains comparable to multiple sequential interventions in fewer interaction rounds, although the benefit varies with the dataset and maximum cardinality K

D.4. Results for different expert costs

The main paper reports results at $\alpha = 0.01$ and visualises the quality–cost trade-off across additional expert costs. Here, we provide the exact values corresponding to the curves shown in the main figure. Each table entry reports the video-level mIoU and average selected cardinality, $(\mathrm{mIoU}, \widehat{k})$, for a given value of α.

D.5. Qualitative results

Figures 9–12 present qualitative comparisons on VOST, CAMotion, SUN-SEG, and VTUS using examples spanning different levels of initial propagation quality. Across these cases, our method selects correction cardinalities and frame sets that achieve higher final IoU and lower system cost than Top-$k(x)$. The examples further illustrate complementary frame selection in difficult videos and the avoidance of unnecessary corrections when the initial propagation is already reliable.

Table 12. Ablation on the expert costs α on VOST and CAMotion. Each entry reports $(\mathrm{mIoU}, \widehat{k})$, where $\widehat{k}$ is the mean number of selected correction frames. Continued in Table 13.

	VOST			CAMotion		
α	**Ours**	**Top-$k(x)$**	**LECODU**	**Ours**	**Top-$k(x)$**	**LECODU**
0	(0.705, 3.97)	(0.679, 4.00)	(0.670, 4.00)	(0.859, 3.99)	(0.850, 4.00)	(0.847, 4.00)
0.005	(0.702, 3.47)	(0.663, 2.93)	(0.670, 4.00)	(0.850, 2.03)	(0.844, 1.77)	(0.843, 2.07)
0.01	(0.699, 2.56)	(0.662, 2.82)	(0.658, 3.21)	(0.847, 1.19)	(0.838, 1.30)	(0.837, 1.28)
0.015	(0.685, 2.34)	(0.649, 2.46)	(0.655, 3.11)	(0.838, 0.72)	(0.830, 0.63)	(0.825, 0.52)
0.02	(0.673, 2.03)	(0.644, 2.26)	(0.652, 2.91)	(0.836, 0.67)	(0.826, 0.47)	(0.820, 0.14)
0.05	(0.622, 1.11)	(0.607, 1.39)	(0.610, 1.40)	(0.817, 0.03)	(0.818, 0.04)	(0.802, 0.00)
0.1	(0.609, 0.97)	(0.554, 0.63)	(0.531, 0.53)	(0.817, 0.03)	(0.815, 0.01)	(0.802, 0.00)
0.15	(0.522, 0.30)	(0.511, 0.34)	(0.476, 0.01)	(0.815, 0.01)	(0.802, 0.00)	(0.802, 0.00)
0.2	(0.484, 0.04)	(0.506, 0.24)	(0.473, 0.00)	(0.815, 0.01)	(0.802, 0.00)	(0.802, 0.00)
0.3	(0.473, 0.00)	(0.479, 0.06)	(0.473, 0.00)	(0.802, 0.00)	(0.802, 0.00)	(0.802, 0.00)
0.4	(0.473, 0.00)	(0.473, 0.01)	(0.473, 0.00)	(0.802, 0.00)	(0.802, 0.00)	(0.802, 0.00)

Table 13. Ablation on the expert costs α on SUN-SEG and VTUS, continued from Table 12.

	SUN-SEG			VTUS		
α	**Ours**	**Top-$k(x)$**	**LECODU**	**Ours**	**Top-$k(x)$**	**LECODU**
0	(0.868, 4.00)	(0.853, 4.00)	(0.849, 4.00)	(0.820, 4.00)	(0.804, 4.00)	(0.780, 4.00)
0.005	(0.866, 2.73)	(0.852, 3.11)	(0.843, 3.03)	(0.820, 3.97)	(0.802, 3.73)	(0.768, 3.64)
0.01	(0.850, 1.71)	(0.842, 1.80)	(0.836, 2.01)	(0.819, 2.81)	(0.795, 2.90)	(0.755, 3.26)
0.015	(0.848, 1.55)	(0.838, 1.71)	(0.828, 1.86)	(0.809, 2.35)	(0.759, 1.90)	(0.737, 2.22)
0.02	(0.847, 1.41)	(0.825, 1.27)	(0.814, 1.46)	(0.798, 2.00)	(0.742, 1.73)	(0.733, 2.14)
0.05	(0.830, 1.00)	(0.803, 0.86)	(0.788, 1.03)	(0.728, 0.90)	(0.717, 0.80)	(0.717, 1.47)
0.1	(0.795, 0.65)	(0.789, 0.68)	(0.755, 0.49)	(0.694, 0.50)	(0.706, 0.53)	(0.652, 0.13)
0.15	(0.768, 0.22)	(0.780, 0.62)	(0.750, 0.46)	(0.676, 0.23)	(0.663, 0.23)	(0.627, 0.00)
0.2	(0.725, 0.03)	(0.770, 0.46)	(0.721, 0.03)	(0.651, 0.07)	(0.660, 0.13)	(0.627, 0.00)
0.3	(0.725, 0.03)	(0.747, 0.16)	(0.711, 0.00)	(0.642, 0.03)	(0.650, 0.07)	(0.627, 0.00)
0.4	(0.725, 0.03)	(0.745, 0.08)	(0.711, 0.00)	(0.633, 0.03)	(0.627, 0.00)	(0.627, 0.00)

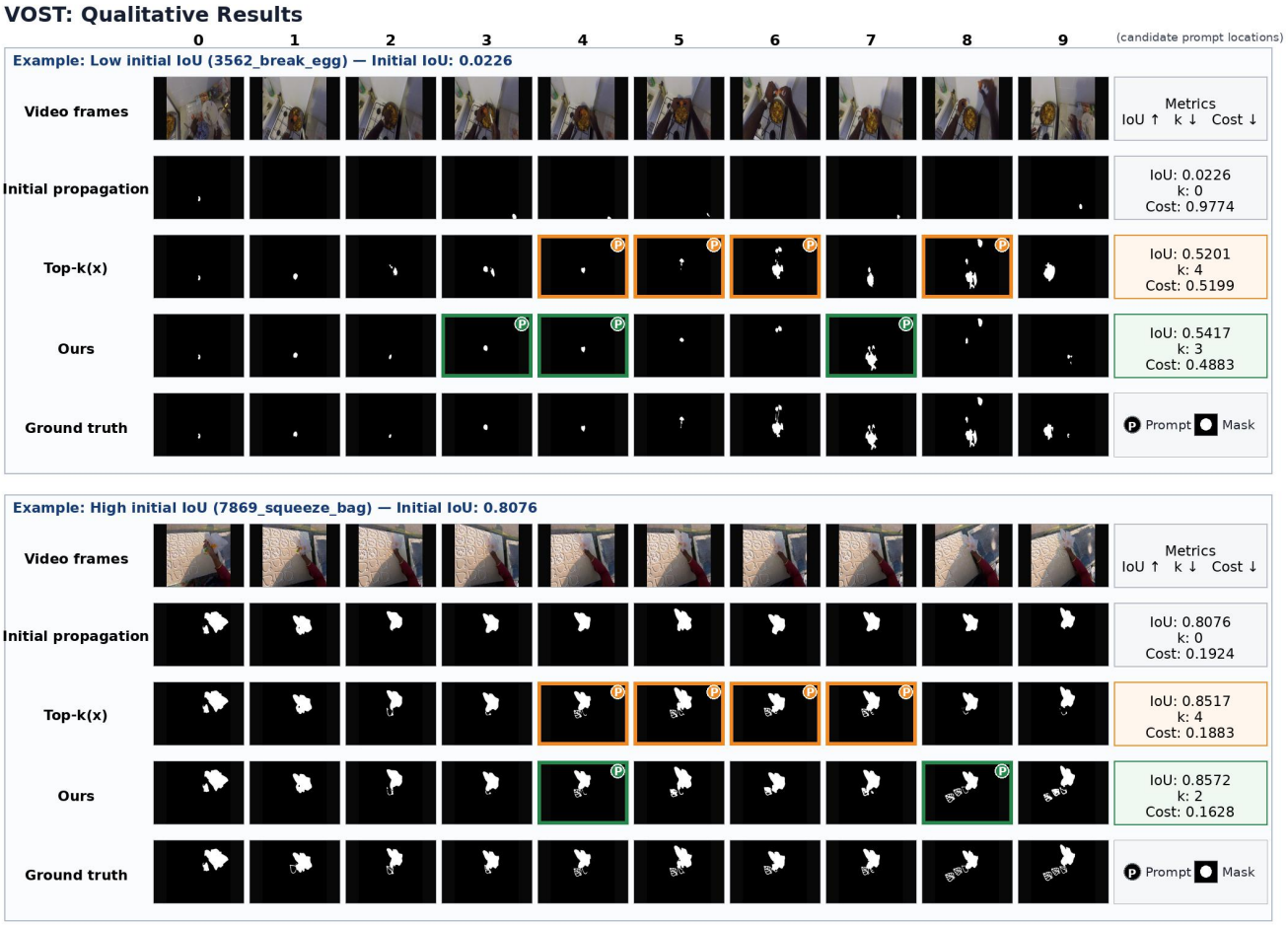


Figure 9. Qualitative comparison on VOST, which contains substantial object appearance and shape transformations. The examples show an egg being broken and a bag being squeezed, where deformation makes mask propagation challenging. The Top-$k(x)$ and Ours rows show the masks obtained after applying all selected corrective prompts and re-propagating the video. In the low-initial-IoU example, our method achieves higher final IoU using fewer prompts and lower system cost. In the high-initial-IoU example, it avoids unnecessary corrections while obtaining slightly higher IoU and lower cost. Coloured borders and prompt markers indicate the selected correction frames.

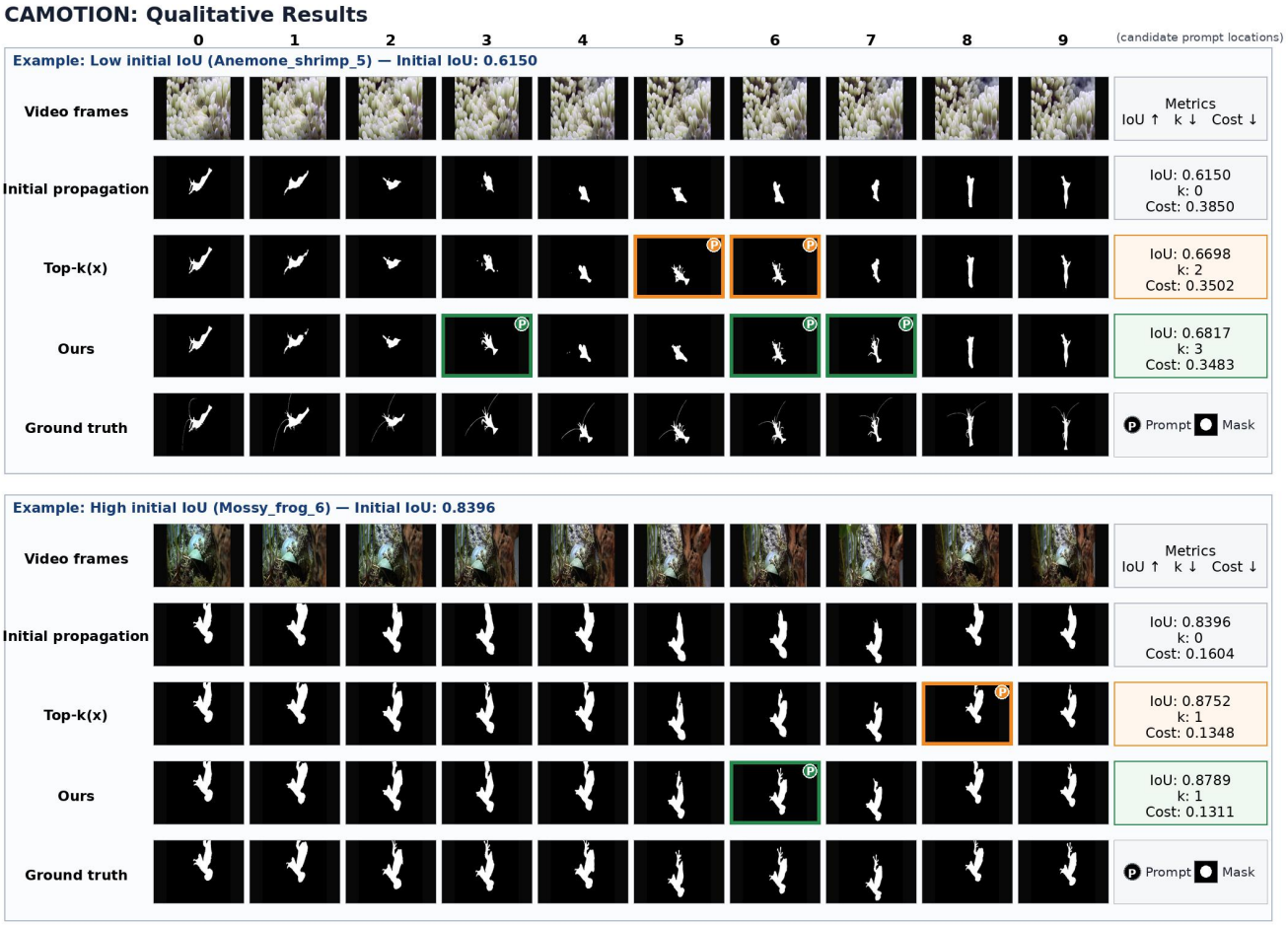


Figure 10. Qualitative comparison on CAMotion, where foreground objects are difficult to distinguish from their surrounding backgrounds. The examples show an anemone shrimp with relatively low initial IoU and a mossy frog with high initial IoU. The Top-$k(x)$ and Ours rows show the masks obtained after applying the selected corrective prompts and re-propagating the video. In the low-initial-IoU example, our method achieves higher IoU and lower system cost by selecting three complementary correction frames. In the high-initial-IoU example, both methods request one correction, while our selected frame yields slightly higher IoU and lower cost. Coloured borders and prompt markers indicate the selected correction frames.

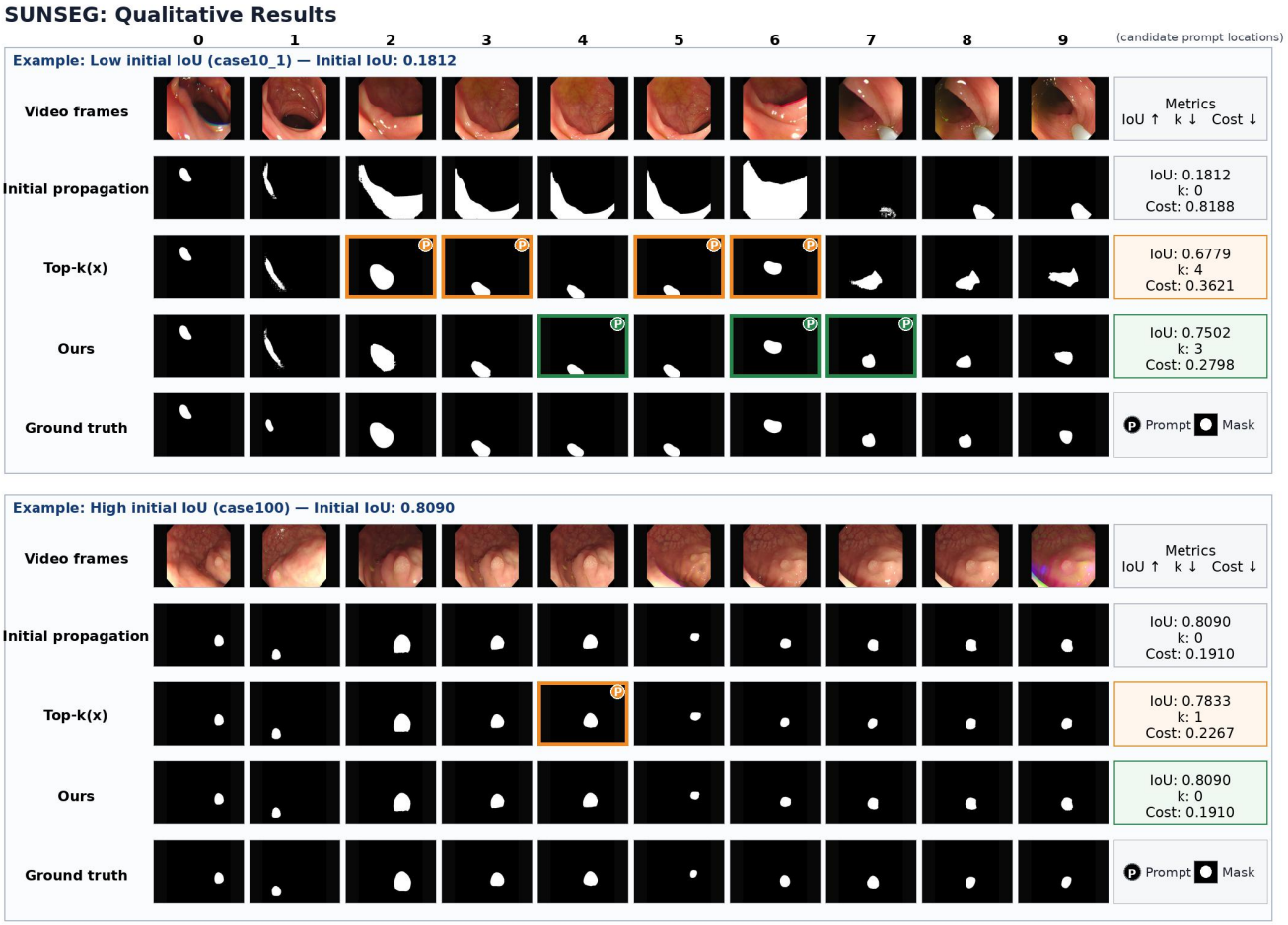


Figure 11. Qualitative comparison on SUN-SEG, where polyp appearance, scale, and viewpoint can vary substantially across colonoscopy frames. The Top-$k(x)$ and Ours rows show the masks obtained after applying the selected corrective prompts and re-propagating the video. In the low-initial-IoU example, our method selects three corrective frames and achieves substantially higher IoU with lower system cost than Top-$k(x)$, which requests four prompts. In the high-initial-IoU example, our method retains the initial propagation with $k = 0$, whereas Top-$k(x)$ requests an unnecessary correction that reduces IoU and increases system cost. Coloured borders and prompt markers indicate the selected correction frames.

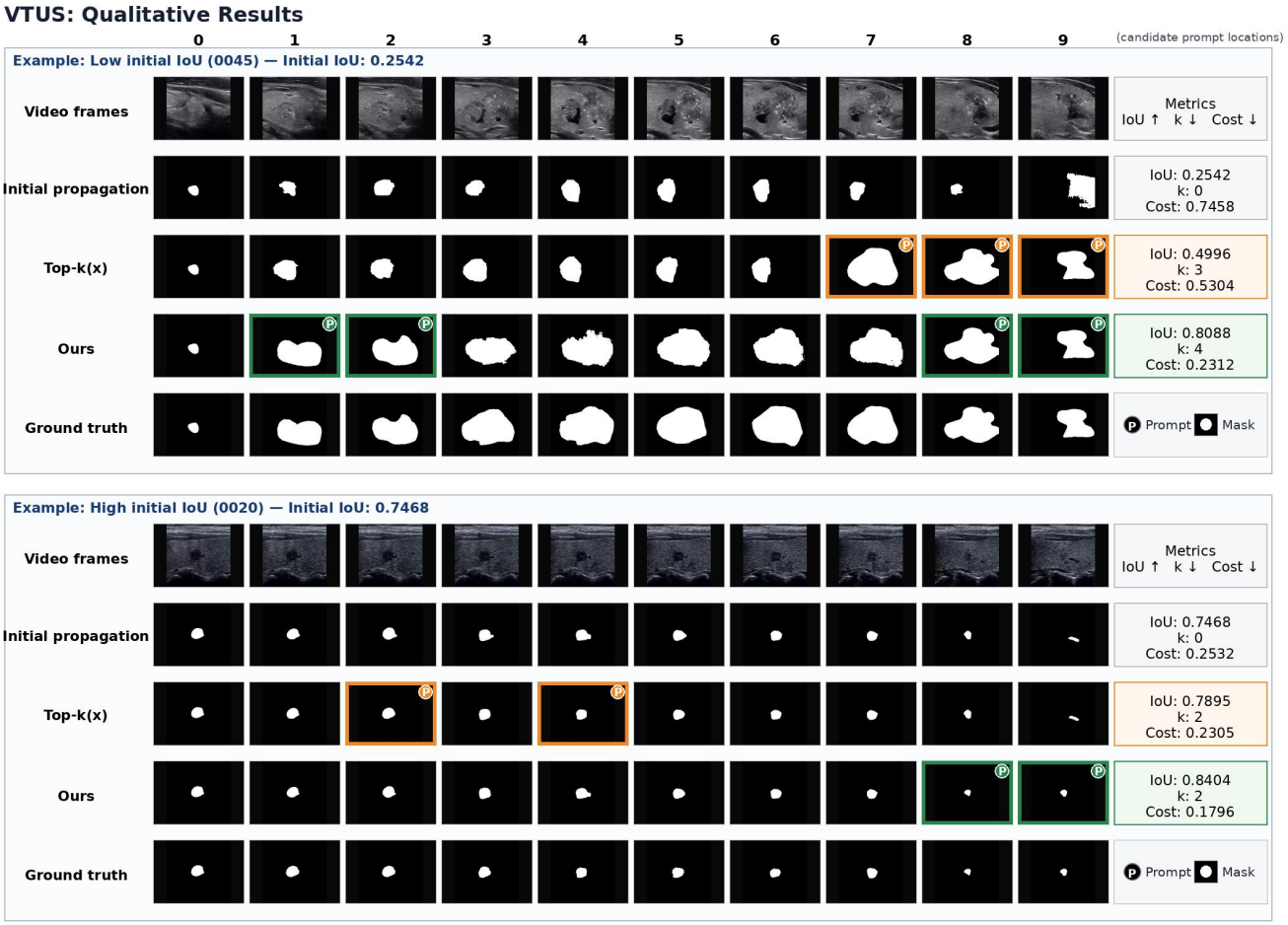


Figure 12. Qualitative comparison on VTUS, where thyroid nodules exhibit low contrast, ambiguous boundaries, and substantial variation in appearance across ultrasound frames. The Top-$k(x)$ and Ours rows show the masks obtained after applying the selected corrective prompts and re-propagating the video. In the low-initial-IoU example, our method selects four complementary correction frames and substantially improves IoU while achieving a much lower system cost than Top-$k(x)$. In the high-initial-IoU example, both methods request two corrections, but our selected frames yield higher IoU and lower system cost. Coloured borders and prompt markers indicate the selected correction frames.